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Digital twin models for architectural heritage conservation

Ming Guo [a,b,c], Xiaoke Shang [a], Jiawei Zhao [e,*], Yanru Shi [a], Mengxi Sun [d], Ming Huang [a,b,c], Guoli Wang [a,b,c], Liqiong Liao [a]

[a] School of Geomatics and Urban Spatial Informatics, Beijing University of Civil Engineering and Architecture, Beijing, 102616, China
[b] Engineering Research Center of Representative Building and Architectural Heritage Database Ministry of Education, Beijing, 100044, China
[c] Key Laboratory of Modern Urban Surveying and Mapping. National Administration of Surveying, Mapping and Geoinformation, Beijing, 100044, China
[d] School of Remote Sensing Information Engineering, Wuhan University, Wuhan, 430072, China
[e] National Marine Data and Information Service, Tianjin, 300171, China

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ABSTRACT

Traditional approaches to architectural heritage conservation often prioritize emergency maintenance, which lacks the capability to monitor and adapt to the dynamic, ongoing changes in structural integrity, thereby compromising long-term conservation objectives. This study introduces a digital twin five-dimensional model framework specifically tailored for architectural heritage conservation, designed to facilitate a shift from reactive to proactive conservation strategies. Incorporating dimensions of physical entity, digital twin, twin data, service, and connection relationships, this framework enables real-time situational awareness and hyper-real-time virtual extrapolation. Through the reconstruction of the historical building information model and the development of both a structural linkage rule model and a displacement trend model based on mechanical properties, the framework offers a comprehensive digital representation from geometric, physical, behavioral, and rule-based perspectives. Validated through a prototype system at the Yingxian Wooden Pagoda in China, a site susceptible to natural and anthropogenic hazards, the findings demonstrate that the digital twin effectively predicts and adapts to spatio-temporal dynamic changes in the pagoda's structure, enhancing resilience and promoting sustainable heritage conservation. This research supports disaster risk reduction strategies, contributing to vulnerability analysis and resilience in heritage conservation amid increasing threats from climate change and other emergent risks.

1. Introduction

As a non-renewable cultural resource, architectural heritage documents the architectural techniques and creative styles of many periods, embodying the cultural implications of social structures, religious beliefs, and lifestyles [1]. Nevertheless, due to variables such as material degradation and human intervention, some architectural historic sites are already compromised [2,3]. They urgently require appropriate conservation and maintenance strategies. Presently, most conservation strategies rely on emergency repair, which

* Corresponding author.
E-mail addresses: guoming@bucea.edu.cn (M. Guo), shangxiaoke333@163.com (X. Shang), zhaojiawei9921@163.com (J. Zhao), 1414349797@qq.com (Y. Shi), mengxisun@whu.edu.cn (M. Sun), huangming@bucea.edu.cn (M. Huang), wangguoli@bucea.edu.cn (G. Wang), liaoliqiong@bucea.edu.cn (L. Liao).

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can guarantee structural integrity in the short term but also pose challenges in preserving originality [4,5]. Consequently, transitioning conservation efforts from passive salvage restoration to proactive preventive maintenance and intervening prior to the onset of threats is essential [6–8].

Digital twin technology [9] offers a novel approach to precisely understanding architectural history by creating a virtual model that represents an entity's complete life cycle state using data and model-driven simulation, monitoring, prediction, and optimization [10, 11]. It can assess the condition of historical structures in real-time using multi-source heterogeneous data and execute hyper-real-time extrapolation, thereby enhancing building health management and minimizing human intervention [12].

The digital twin concept was initially introduced by Michael Grieves in 2003, characterized as a three-dimensional representation of entities, voxels, and their informational interconnections [13]. Subsequently, Negri et al. expanded this concept, highlighting the real-time connection between physical and virtual systems via data synchronization [14]. As research advances, the three-dimensional model is increasingly inadequate to fulfil complicated requirements. Tao Fei and colleagues developed a five-dimensional architecture encompassing physical entities, virtual entities, twin services, twin data, and connectivity linkages [15]. This design has significantly advanced the utilization of digital twin technologies across various study domains.

Although digital twin technology has shown unique advantages in many fields, its application in architectural heritage conservation is still in the exploratory stage, especially in preventive conservation, and the theoretical construction and empirical analysis have not been systematized yet. Vuoto et al. explored its potential in conserving the structural integrity of architectural heritage but lacked the verification of its practical application [16].Tan et al. combined laser scanning with tilt photogrammetry tools for architectural archaeology, but this approach is limited to specific heritage types [17]. There is no universal method for building digital twins of architectural heritage.

Digital twins, as virtual entities, are digital representations of physical entities for understanding, predicting, optimizing and controlling natural objects. The universality of its construction method directly affects the application effect of the digital twin solution. In the face of complex building structures and variable environments, how to construct pervasive digital twins to realize real-time monitoring and early warning is a vital issue in current research [18]. Bordeleau et al. explored various model-driven engineering techniques used in the model building process, identified as potentially overcoming the challenges associated with digital twin modeling [19]. Concurrently, Rasheed et al. provide a comprehensive overview of the methods and techniques currently applied in constructing digital twin models [20]. They discuss the challenges encountered during the construction process in detail, offering specific recommendations and insights from the authors. Funari et al. proposed a parametric Scan-to-FEM framework and validated the application of this method in historical masonry structures through the São Tocato Church in Portugal. However, the applicability to complex or irregular structures is limited [21]. Jouan et al. proposed a digital twin construction method based on the Historical Building Information Model (HBIM) to integrate multidimensional data to monitor changes and propose solutions [22]. The model (HBIM) digital twin construction method integrates multidimensional data to monitor changes and propose solutions. HBIM can accurately characterize complex geometrical structures and physical attributes of architectural heritage by integrating time dimensions and multilevel information [23]. However, most current HBIM applications only serve as static models, failing to capture the dynamic processes of architectural change over time and multifactors and achieving proper preventive conservation. Colucci et al. investigated the interoperability of HBIM with 3D GIS, but their aim was limited to providing more comprehensive data documentation for heritage conservation [24].

This work aims to thoroughly investigate the methods for generating digital twins of architectural heritage utilizing the HBIM model, focusing on the four essential characteristics of "geometry-physics-behavior-rule." This will encompass the intricacies of architectural heritage in all dimensions, including structural modifications, physical conditions, behavioral responses, and adherence to regulations. Consequently, genuine preventive protection can be achieved, offering a novel perspective for architectural asset health monitoring and life cycle forecasting.

2. Methodlogy

2.1. Digital twin five-dimensional model

The digital twin model of architectural heritage is an extensive data-driven framework designed to integrate the multi-physical, multi-scale, and multi-probability simulation processes throughout the entire life cycle of architectural heritage. It aims to comprehensively capture data regarding the dynamic changes of architectural heritage over time and under various influences through multi-level real-time/quasi-real-time situational awareness and ultra-real-time virtual extrapolation, facilitating the transition from salvage to preventive conservation. Architectural heritage preventive conservation is a critical concern based on the notions above. Based on the above concept, for the needs of preventive protection of architectural heritage, this study combines the existing digital twin five-dimensional model to give the framework of the digital twin five-dimensional model of architectural heritage, which is visualized in the expression of Equation (1)

$$M_{HBDT} = \{HBPE, HBDT, HBDD, HBSs, HBCN\} \tag{1}$$

In the given formula, M_{HBDT} denotes the five-dimensional model of architectural heritage digital twin; *HBPE* represents the physical entity of architectural heritage; *HBDT* represents the digital twin of architectural heritage; *HBDD* represents the data of architectural heritage twin; *HBSs* represents the intelligent monitoring and diagnostic service of architectural heritage; and *HBCN* signifies the connectivity relationship between different dimensions, as illustrated in Fig. 1.

Historic Building Physical Entity (*HBPE*) represents the objectively existing architectural heritage and the deployed sensors, with significant hierarchy and interdependence. Historic Building Digital Twin (*HBDT*) refers to the actual mirror model of the physical entity created by digitalization, which accurately portrays the physical entity of the architectural heritage in "multi-dimension, multi-perspective, multi-width and multi-granularity" and is responsible for obtaining and feeding back the data information of the physical entity of the architectural heritage. Historic Building Digital Data (*HBDD*) is mainly composed of architectural heritage physical entities, digital twins, services and other data, and is the driver and core of architectural heritage digital twin. Historic Building Smart monitoring and diagnostic Services (*HBSs*) is used to realize digital management and early warning of architectural heritage status information by integrating data, algorithms and results of the digital twin five-dimensional modelling framework and analyzing the monitoring data using embedded algorithms to promote its preventive protection. Historic Building Connection Network (*HBCN*) is used to realize the information circulation between the architectural heritage digital twin five-dimensional model framework. Synchronizing the interaction of the central triangular part (*HBSs*, *HBPE*, *HBDT*) and maintaining a high degree of consistency at the data level with the middle part (*HBDD*) makes the middle part drive the operation of the whole framework.

2.2. Digital twin modeling

Architectural heritage digital twin construction is the digital modeling of physical entities' properties, methods, behaviors, and other characteristics in a digital space to simulate and predict their state and behavior. In order to ensure that the twin is highly consistent with the physical entity, it needs to be modeled in a virtual environment from the fusion of four dimensions: geometry, physics, behavior and rules, as shown in Fig. 2.

(1) Geometric Modeling

Geometric modeling (M_G) is a digital representation of architectural heritage, detailing the geometric attributes of elements such as columns, beams, and sapphires. These define the heritage's intricate internal and external structures and constitute the initial phase in creating a digital twin. Point cloud modeling and building information modeling are commonly utilized. Point cloud modeling uses laser scanning and other techniques to gather dense point cloud data, thereby capturing the intricate aspects of the item. The volume of data remains substantial and necessitates high-performance computing resources. This work employs a hybrid point cloud approach and building information modeling to ensure data consistency to reconstruct physical things' geometric form and non-graphical data, thereby establishing a foundation for future model analysis and interactive functions.

(2) Physical Modeling

Physical modeling (M_P) is the addition of physical properties and constraints to a geometric model to simulate the dynamic response of built heritage under natural environments (e.g., wind, rain, earthquakes, etc.) or anthropogenic forces, such as stresses and deformations. Mathematical approximations are usually made using finite element modeling techniques to construct models reflecting the physical phenomena of built heritage at macro and micro scales. Training and optimizing the models with real-time data streams can achieve a high-level mapping of the built heritage's physical entities.

(3) Behavioral model

The behavioral model (M_B) mainly describes the real-time behaviors of physical entities under the coupling of the external environment and internal causes in different time and space scales, such as evolutionary behaviors, performance degradation be-

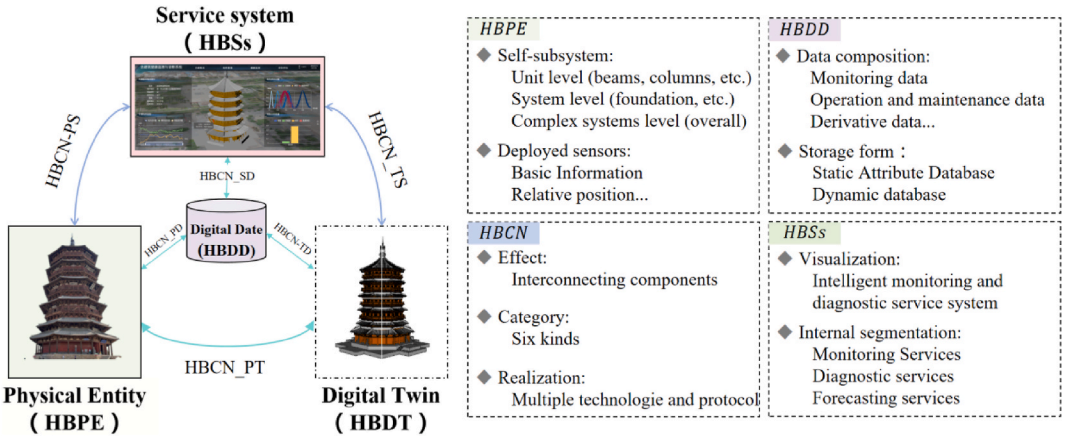


Fig. 1. Organizational chart of the five-dimensional model framework of the architectural heritage digital twin.

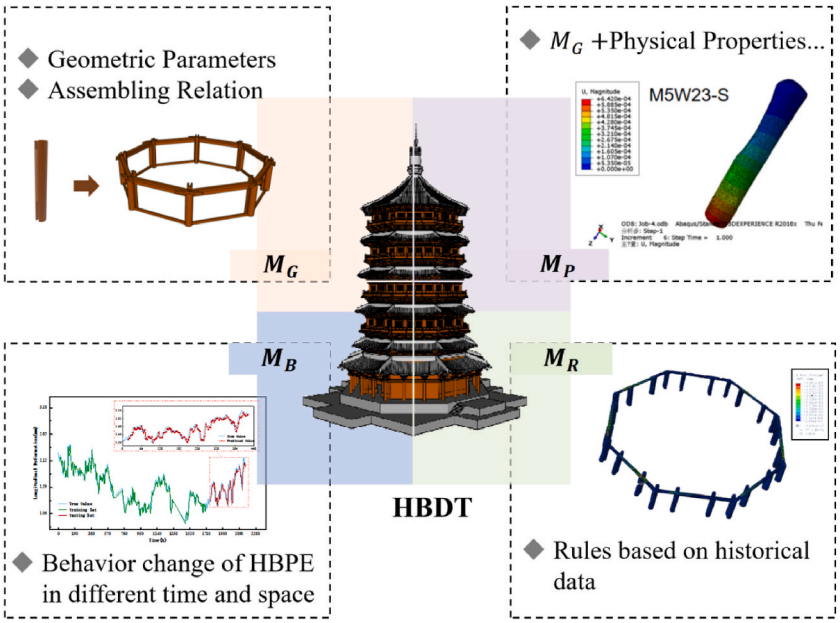


Fig. 2. Basis of digital twin model construction.

haviors, etc., as time advances. Constructing a behavioral model includes defining the purpose and application scenarios, collecting relevant data as necessary inputs, selecting appropriate modeling methods, and performing validation and calibration. The model must be continuously updated and optimized based on new data.

(4) Rule Model

The rule model (M_R) delineates regulations derived from historical architectural heritage data, encompassing data correlation, generalization of implicit knowledge, and design specifications for wooden structures. The development of the rule model may depend on the synthesis of prior experiences or sophisticated data analytics technologies, like machine learning and data mining, to perpetually extract updated rules automatically.

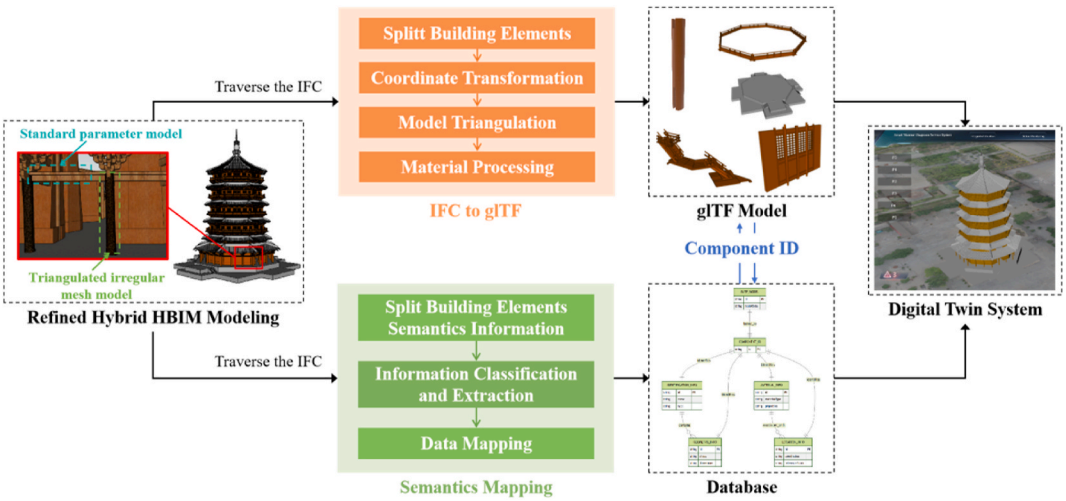


Fig. 3. HBIM-based geometric and physical model construction.

3. Related work

3.1. HBIM-based geophysical modeling

Historical Building Information Modeling (HBIM) is a specialized modeling technique designed for architectural heritage, adept at accurately representing the intricate geometry of such structures while encompassing multidimensional data, including physical properties and material characteristics. This study integrates a conventional parametric model and an irregular triangular mesh model to precisely represent the geometric distinctiveness of built heritage in HBIM modeling. This study proposes a three-step digital-model decoupling, geometric conversion, and attribute mapping process to enhance the weak data interoperability between HBIM and the established digital twin framework integration scheme, ensuring that the HBIM model can be rendered and interacted with within the digital twin framework. The procedure is illustrated in Fig. 3.

Fine modeling is a core aspect of architectural heritage digitization. In order to balance high-quality digital representation and efficiency, this study blends two approaches: standard parametric modeling and participatory regular triangular mesh modeling. Fine hybrid modeling integrates a specific component's irregular triangular mesh model into the standard parametric model framework. In order to avoid the problem of missing data caused by cutting point clouds during irregular triangular mesh modeling, the irregular triangular mesh model is finely tuned using the standard model parameters. The irregular triangular mesh model is precisely positioned and moved to the predetermined position of the standard parametric model to realize the accurate spatial docking of the two. The exemplary hybrid model of the Yingxian wooden tower is shown in Fig. 4. After completing the model fusion, attribute input and Industry Foundation Classes (IFC) type categorization are performed, and the model is exported in IFC format.

Due to their intricate architecture and attribute data, IFCs are inappropriate for direct rendering and cannot be used for graphical representation in a 3D format. Consequently, IFC files must be transformed into a compatible format. Gl Transmission Format (glTF) encompasses all elements necessary to create a 3D scene. glTF serves as a target format for converting IFC model files to finalize the building of geometric models within the defined digital twin architecture.Nonetheless, the straight conversion from IFC to glTF format is more intricate because of existing technical constraints. The OBJ format is a prevalent standard for 3D model files, encompassing vertex coordinates, usual coordinates, and texture coordinates. This study selects OBJ as an intermediary to develop the IFC-OBJ-glTF conversion framework to create glTF model files. The essence lies in processing the relative coordinate system of HBIM during the conversion of IFC-OBJ to the global coordinate system of WebGL, aiming to accurately map the spatial relationships of geometric information. The coordinate conversion procedure encompasses several essential steps: Initially, the local point coordinates of the components must be extracted from the IFC file; subsequently, these local coordinates are integrated with the global coordinates of the overall model by iteratively applying the matrix transformation layer by layer in the following equation to derive the absolute coordinates of the vertices ultimately. The coordinate transformation is presented in Equation (2).

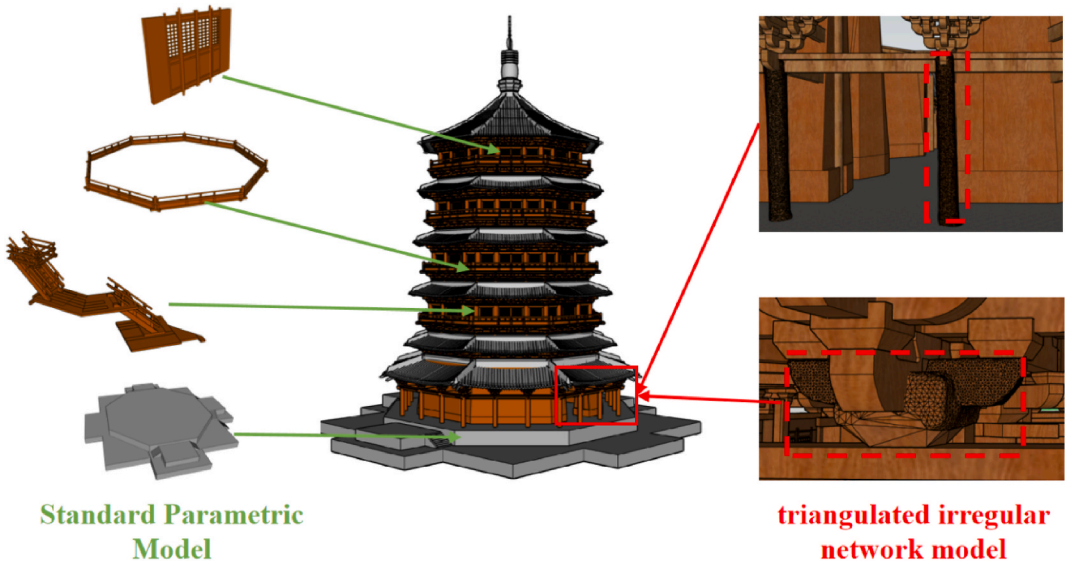


Fig. 4. Yingxian wooden pagoda fine mix HBIM modeling.

$$
\begin{aligned}
V_{\text{Model}_n} &= (\mathrm{T}_{n-1} \times R_{n-1} \times S_{n-1}) \times V_{\text{Component}_{n-1}} \\
&\mathrm{T} \\
&= \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
&R \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha & 0 \\ 0 & -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\beta & 0 & -\sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} \cos\gamma & \sin\gamma & 0 & 0 \\ -\sin\gamma & \cos\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
&S \\
&= \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
\end{aligned} \tag{2}
$$

The subscript $n = 1, 2, \cdots, m$ represents the number of iterations of the component in the hierarchical model. The subscript $V_{\text{Component}_{n-1}}$ represents the coordinate position in the local coordinate system of level $n-1$. The subscript V_{Model_n} represents the coordinate position in the coordinate system of the model at level n. The translations of the x, y, and z axes are represented by t_x, t_y, and t_z, respectively. The angles of rotation of the x, y, and z axes, with a positive counterclockwise direction specified, are represented by α, β, and γ, respectively. The scaling multipliers are represented by s_x, s_y, and s_z.

OBJ-glTF conversion entails analyzing OBJ and MTL files to retrieve geometric and material data. Geometric data is transformed into appropriate spatial structures, while material information is translated into visual qualities. Scene and node structures represent the model's spatial configuration, while buffers and accessors are arranged to enhance data storage efficiency. Optimization techniques are employed to compress data, and all transformed data is systematically arranged and stored in glTF files.

During the attribute mapping phase, it is essential to verify the completeness and precision of the attribute information derived from the IFC file and its adaptability and scalability to fulfill digital twin requirements. This study employs a relational database for attribute mapping, constructs the attribute table through IFC semantic analysis classification, extracts attributes by parsing a JSON file for storage in the database, and establishes bi-directional links between the attributes and the glTF model using unique identifiers to ensure synchronized data updates. Attribute information is delineated by attribute and relationship entities, which construct the

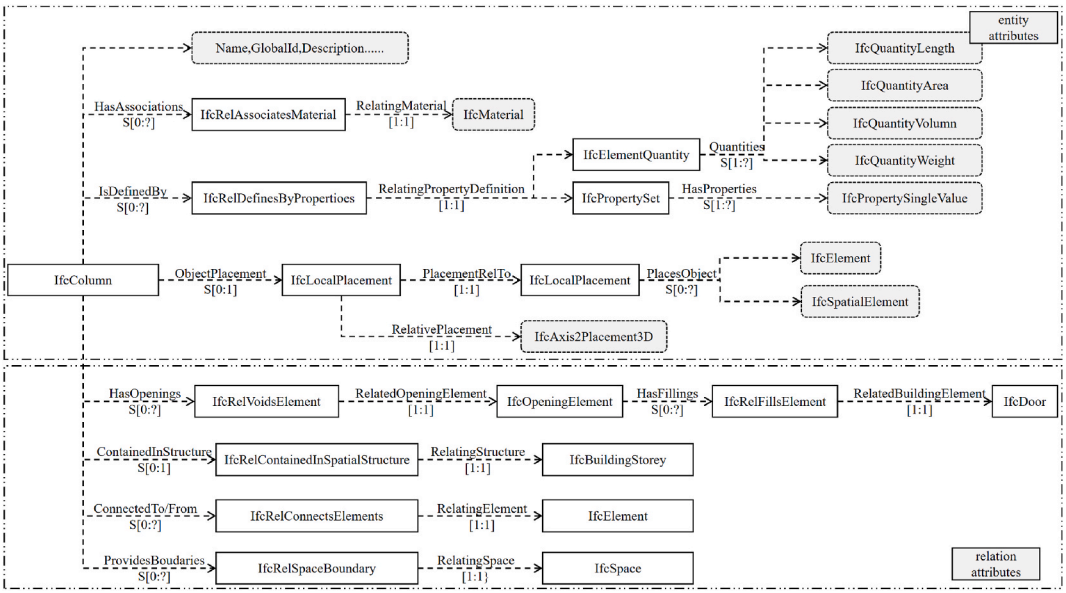


Fig. 5. Building Element Attribute Element Extraction (IfcColumn as an example).

associative relationships between objects. The non-geometric attribute information extraction procedure is exemplified using IfcColumn in IFC, as depicted in Fig. 5.

3.2. Modeling of structural linkage rules taking into account mechanical properties

In digital twin modeling, a rule model delineates the regulations governing the interaction and data synchronization between physical entities and their digital counterparts. It encompasses mapping data structures and properties, as well as regulations for behavior, operations, and responses. In structural linkage rule modeling that considers mechanical properties, the goal is to ensure that the digital twin system not only reflects the geo metric information of the structure but also accurately simulates the structure's response under various mechanical loads and environmental conditions.

Following geometrical-physical modeling to precisely represent the structural components, Finite Element Analysis (FEA) is employed to develop a mathematical model of the structure, account for the linkage effects, and examine the forces among the components. The geometric model is partitioned into finite element meshes. Finite element simulations are initially conducted to examine the mechanical behavior by assessing the stresses, strains, and displacements of the parts under diverse loading conditions. Subsequently, structural linkage analysis is conducted to examine the interactions among the different components of the structure and to assess the impact of the linkage effect on the overall structural response.

Mechanical properties denote the characteristics and behavior of structural materials and components under external forces. These qualities are crucial when evaluating and designing structures, as they directly influence the structure's stability, durability, and safety. Specify the material parameters, including the modulus of elasticity and Poisson's ratio, for all structural components. This study focuses on a timber tower's single-story, outwardly fluted column structure as the primary subject. The outer slotted column network is a type of larch; the wood exhibits orthogonal anisotropy; the columns endure vertical loads, primarily from longitudinal grain compression; appendages and other horizontal connecting elements experience mainly transverse grain compression, with their material properties referenced in Table 1. The modulus of elasticity encompasses all material qualities, including the modulus of elasticity and Poisson's ratio. The modulus of elasticity is defined within the elastic range as the ratio of tensile or compressive stress to strain per unit length of the material. Conversely, Poisson's ratio quantifies the extent to which a material contracts in the perpendicular direction when subjected to tensile stress in a single direction. It is the negative of the ratio of transverse strain to longitudinal strain.

The numerical simulation of the regular model obtained a series of deformation values under the consideration of long-term loading only, and the numerical simulation of the fictitious model obtained a series of deformation values under the consideration of long-term loading and aging to explore the relationship between the static stiffness reduction function brought about by the natural aging effect. In the frontal square at the midpoint, the column and frontal square connection selected several characteristic points, the points of, and static stiffness to establish a relationship between them. In this study, the variation rule of the static stiffness of wooden columns with service time is shown in Equation (3). According to the form of the load-deformation function of the column under long-term self-weight load; it is determined that the deformation of the wooden column is inversely proportional to the static stiffness (EI).

$$\begin{cases} E_z{}'I_z{}' = kE_zI_z \\ k = e^{at_z} \\ E_z{}'I_z{}' = kE_zI_z = e^{at_z}E_zI_z \end{cases} \qquad \frac{y_z}{Y_z} = \frac{E_z{}'I_z{}'}{E_zI_z} = \frac{e^{at_z}E_zI_z}{E_zI_z} = e^{at_z} = k \tag{3}$$

Where: k is the static stiffness reduction coefficient of the wooden column; t_z is the service time of the wooden column member, unit: years; a is the reduction constant; E_zI_z represents the static stiffness of the wooden column in the initial state; $E_z{}'I_z{}'$ represents the static stiffness of the wooden column after considering the reduction effect of natural aging; y_z represents the deformation of the wooden column under the long-term load; Y_z represents the deformation of the wooden column under the joint action of long-term load + natural aging.

The joint nodes of the column network are primarily connected using mortise-and-tenon joints, treated as semi-rigid connections. Mortise-and-tenon connections are a type of semi-rigid connection with unique variable stiffness characteristics, which are neither ideal hinges (free rotation) nor fully rigid connections (no relative rotation). The moment-rotation relationship of a typical through-tenon joint exhibits a bilinear characteristic, which can be described by the following equation (4):

Table 1
Material parameters.

Material	Elastic Modulus/MPa			Poisson's Ratio		
	E_x	E_y	E_z	μ_{yz}	μ_{xz}	μ_{xy}
Vertical component	1000	275	650	0.3	0.02	0.035
Horizontal component	275	1000	650	0.02	0.3	0.035

$$M(\theta) = \frac{k_0(\theta)\theta}{\left(1 + \left|\frac{\theta}{\theta_0}\right|^n\right)^{\frac{1}{n}}} + M_{friction} \tag{4}$$

Where k_0 is the initial rotational stiffness, θ_0 is the reference rotation angle, n is the shape control parameter, $M_{friction}$ is the frictional damping term, θ is the relative rotation angle of the joint node, and M is the moment at the joint.

The second-floor columns are linked to the lower auxiliary layer by "forked columns." The foot nodes are combined to account for limitations related to medical treatment and are examined using two-way finite element analysis. Fig. 6 illustrates the finite element model of the outer slot column network and the corresponding analytical method.

3.3. Behavioral model construction based on long and short-term memory networks

The behavioral characteristics of architectural heritage are complex and diverse, particularly in how structural components respond to external changes. Given the central role of wooden columns in maintaining the structural stability of such heritage, this study focuses on the behavioral analysis of wooden columns, exploring effective methods to capture and simulate the dynamic displacement changes of these critical structural elements over time. Displacement serves as the most direct, quantifiable, and easily monitorable external indicator of structural performance and potential damage. It intuitively reflects changes in structural loading conditions, joint relaxation, component damage, or foundation settlement, making it a crucial early-warning metric for assessing both the immediate health status and long-term evolutionary trends of a structure. The rule-based model transforms the complex physical mechanisms of force transmission between components into a displacement-driven modeling framework. This transformation ensures strong consistency among the model's inputs, internal states, and outputs in terms of physical meaning. Therefore, selecting deformation displacement as the prediction target enhances the model's interpretability and intuitiveness.

When selecting a model foundation, although traditional predictive models such as linear regression, grey prediction, Markov prediction, traditional time series analysis, and neural networks are widely applied across multiple fields, not all models are suitable for the research subject of this paper. The linear regression model assumes a linear relationship between the dependent variable and the independent variables, and that the independent variables are linearly independent, thus unable to handle nonlinear relationships; Grey prediction is primarily used for handling incomplete information in sequence data and performs poorly in predicting nonlinear data; Markov prediction is suitable for predicting the probability of state transitions but has limited direct predictive capability for non-stationary or continuous data; most traditional time series analysis methods, such as ARIMA and VAR, are based on linear assumptions and have limited ability to capture nonlinear dynamics. In deep learning models, the Backpropagation (BP) neural network, as a representative of traditional neural networks, does not allow neurons within the network layers to exchange information, which limits the network's understanding of the semantic relationships between sequential data. In contrast, traditional recurrent neural networks (RNNs) add inter-layer node connections to BP neural networks, allowing information to be transmitted within sequences and combining historical information with the current moment. However, due to the gradient multiplication in RNNs, they face issues of gradient vanishing or explosion when handling long sequence data, limiting their predictive period. To address these issues, Long Short-Term Memory (LSTM) neural networks introduce an internal state structure within units to preserve the current state and propagate it to the next time step, while selectively controlling the flow of information through gating mechanisms. These features enable LSTMs to excel in handling complex time-series data requiring the capture of long-term dependencies. Additionally, LSTMs support both univariate and multivariate predictions, making them adaptable to any scenario. Therefore, the LSTM model is particularly suitable for addressing the long-term sequence prediction problem of displacement in the column heads of ancient architectural wooden columns in this study. Furthermore, establishing a digital twin of ancient architecture requires a comprehensive understanding and prediction of the behavior of wooden columns, including interactions between different directions of load. Thus, this study constructs a multivariate-to-multivariate network based on the LSTM model.

3.3.1. Data cleaning

The data must meet the equal interval criterion for time series modeling and analysis. Consequently, it is essential to extract the requisite data from the raw data at uniform intervals to create a dataset. The chosen dataset originates from displacement monitoring

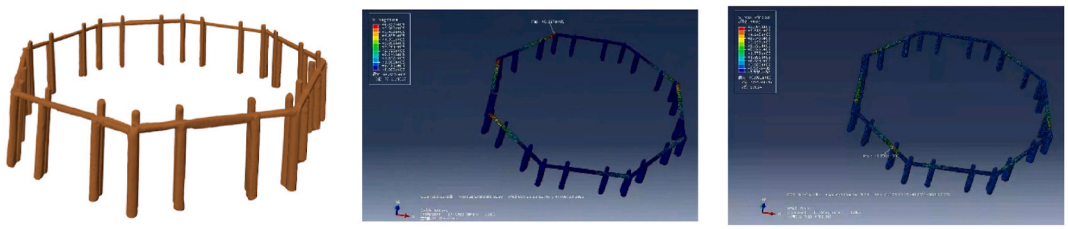


Fig. 6. The finite element model and analysis process of the external groove column network.

data collected between July 1, 2023, and October 12, 2023, at the Yingxian Muta in China. This data encompasses displacements in the longitudinal, transverse, and vertical axes (measured in millimeters). The monitoring robot is equipped with high-precision measurement capabilities, offering an accuracy of 0.5". It operates reliably in outdoor environments, ensuring both the accuracy and consistency of data collection. Measurements are taken every 4 h, resulting in a time series dataset with 624 equally spaced observations. The data collection frequency is determined based on multiple factors, including the rate of structural changes and the robot's power consumption. Displacement and deformation in wooden structures occur gradually over time and are influenced by environmental conditions. Therefore, the chosen frequency effectively captures structural changes without causing unnecessary resource consumption.

The collected data is wirelessly transmitted in real time to the monitoring cloud platform for storage and subsequent processing. This wireless transmission approach effectively reduces latency, ensuring the continuity and timeliness of the data stream. However, network signals may be affected by environmental interference, transmission distance, or weather conditions, which can lead to packet loss or transmission interruptions, resulting in missing values within the time series data. To maintain data quality and accuracy, appropriate data imputation methods must be employed to fill these gaps. The prevalent techniques for addressing missing values are classified into three categories: direct rounding, imputation with predetermined values (such as sample mean, median, or mode), and proximity interpolation (including linear interpolation and k-nearest neighbor interpolation). This study excludes the direct rounding method due to the necessity of analyzing the time-series data from each monitoring point. Utilizing fixed values to supplement the monitoring data would diminish the overall randomness of the data, thereby rendering this approach inadvisable. Given the superior data continuity and minimal missing values at the monitoring points, the linear interpolation method is employed for value imputation, as indicated in Equation (5). This method presumes a linear change between data points, so estimating the missing values between two known data points and effectively preserving the continuity and trend of the data.

$$V_{interpolated} = V_{start} + \frac{(V_{end} - V_{start}) \times (T_{interpolated} - T_{start})}{(T_{end} - T_{start})} \tag{5}$$

Where $V_{interpolated}$ denotes the interpolated data point, while V_{start} and V_{end} represent the known data points before and after interpolation, respectively. Additionally, T indicates the matching time point.

The LSTM model is sensitive to the scale and range of the data; therefore, it is essential to normalize all features to a uniform range, typically [0,1] or [−1,1]. This process accelerates model iteration, enhances convergence speed during training, and minimizes model error. This study employs the Min-Max scaling technique to normalize the data set ($(x_1, x_2, x_3, \ldots, x_n)$), transforming it following Equation (6) to reside inside the interval [0,1].

$$y_i = \frac{x_i - \min}{\max - \min} \tag{6}$$

Where min refers to the lowest value in the sample series data, whereas max refers to the highest value in the sample series data.

3.3.2. Model construction

Although a single LSTM unit can effectively address the gradient vanishing and explosion problems inherent in traditional RNN models, its predictive accuracy remains limited by the complexity of the network architecture. Increasing the stacking depth of LSTM layers can significantly enhance the model's capability to process long-sequence data. In this study, a three-layer stacked LSTM architecture is employed, designed based on the core principle of progressively increasing feature representation complexity in deep neural networks. Following careful evaluation, three LSTM layers were selected, consisting of 100, 200, and 300 neurons, respectively. The initial layer captures fundamental temporal patterns from the original input through finite-capacity focusing. Its 100-unit configuration ensures the fidelity of local feature extraction while minimizing the risk of overfitting due to redundant parameters in the lower layers. The intermediate layer serves as an integration hub for cross-period dependencies, with an expanded capacity of

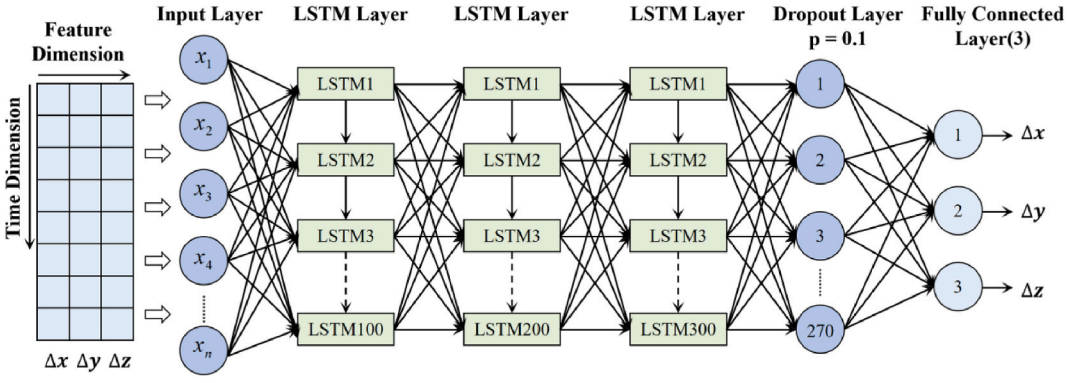


Fig. 7. LSTM-based model network for displacement behavior prediction.

200 units, which significantly enhances the network's nonlinear representation capabilities. This increment was systematically determined through hyperparameter tuning as the optimal balance between gradient stability and model expressiveness. The top layer, with 300 units, represents the engineering-optimal configuration, derived from the convergence of three key constraints: task complexity, data scale, and computational resources. Specifically, the maximum time dependency length required for the modeling task demands a theoretical minimum of 250 units; the upper limit of model capacity, as supported by the training sample size, is approximately 346 units; and the maximum feasible dimension under hardware memory constraints is 320 units. The intersection of these constraints defines a feasible solution space centered around 300 units. The entire network and internal structure are shown in Fig. 7 below.

To meet the training requirements of stacked LSTM models, this study employs the Adaptive Moment Estimation (Adam) algorithm as the optimization method. Adam integrates the benefits of momentum-based methods and adaptive learning rate adjustment, enabling the model to effectively capture the high-dimensional nonlinear characteristics of architectural heritage displacement data. This facilitates efficient convergence on complex loss surfaces while helping to avoid entrapment in local minima.

3.3.3. Hyperparameter settings

To optimize the performance of the LSTM deep learning model for displacement prediction tasks, this study carefully controlled and selected several key hyperparameters, including the learning rate, look_back (the time step length for sequence prediction), batch_size (the batch size), and epoch (the number of training iterations). Among the three displacement directions—longitudinal, lateral, and vertical—vertical displacement is generally easier to measure and exhibits higher measurement accuracy. Therefore, the Mean Absolute Error (MAE) of vertical displacement in the prediction results is used as the primary evaluation metric for assessing the effectiveness of different hyperparameter configurations.

(1) Learning rate

Although the Adam adaptive optimization algorithm can automatically adjust the learning rate during training, the choice of the initial learning rate still significantly affects the model's training dynamics and final performance. To evaluate this impact, other parameters were held constant while different initial learning rates were tested. As shown in Fig. 8, the model's MAE gradually decreases as the initial learning rate increases up to 0.001, indicating an improvement in prediction accuracy. However, once the learning rate exceeds 0.001, the MAE begins to rise, suggesting a decline in performance. Based on this observation, the model achieves its highest accuracy when the initial learning rate is set to 0.001. Therefore, this value is selected as the optimal initial learning rate for subsequent experiments.

(2) Look back

The choice of time step length (look_back) directly influences the model's ability to capture long-term dependencies and patterns in time series data, thereby affecting the accuracy and reliability of predictions. By fixing other parameters, a comparative analysis was conducted using different time step lengths. Fig. 9 illustrates how the model's MAE varies with look_back values ranging from 1 to 10. As shown in the figure, the MAE fluctuates across different time step lengths. Notably, when the look_back is set to 3 and 5, the model achieves lower MAE values, indicating higher prediction accuracy. In contrast, higher MAE values at look_back values of 2 and 9 suggest reduced model performance. These fluctuations may be attributed to the model's varying ability to capture temporal dependencies at different time window lengths. Based on this trend analysis, a look-back of 5 is identified as the optimal choice for predicting the displacement time series at monitoring point No. 223.

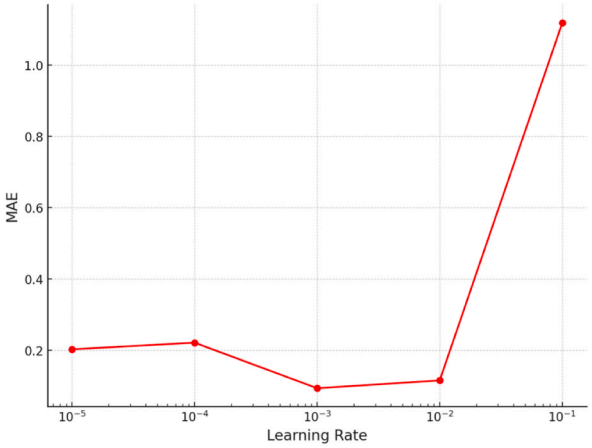


Fig. 8. Effect of learning rate on model accuracy.

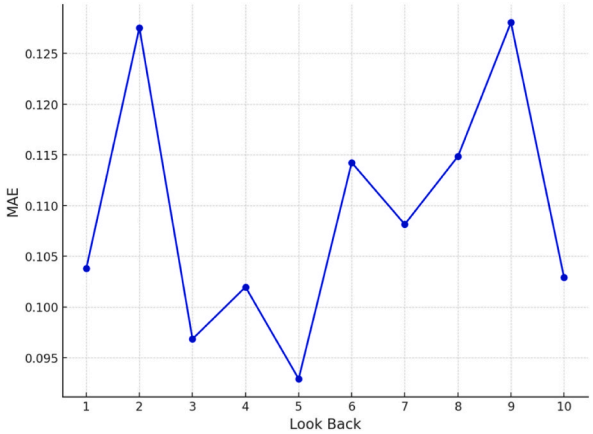


Fig. 9. Effect of look back on model accuracy.

(3) Batch size

The batch size determines the amount of data the model processes in each training iteration, influencing both learning efficiency and memory usage. With all other parameters held constant, different batch sizes—ranging from 1 to 512 in powers of two—were tested to observe their effect on model performance. As shown in Fig. 10, the model's MAE initially decreases and then increases as the batch size grows. Specifically, when the batch size is small (between 1 and 16), the MAE remains low, indicating strong model performance. However, as the batch size increases beyond 32, the MAE begins to rise gradually, suggesting a decline in prediction accuracy. It is worth noting that smaller batch sizes, while improving performance, can significantly increase training time. Therefore, to strike a balance between training efficiency and model accuracy, a batch size of 8 or 16 is recommended, with further adjustments made based on the specific task requirements.

(4) Epochs

Each training epoch consists of one or more forward and backward propagation processes, allowing the model to learn from the training data and adjust its weights. By fixing other parameters and varying only the epoch size, Fig. 11 shows how the Mean Absolute Error (MAE) changes with the number of training epochs (from 50 to 500). During the early stages of training (50–150 epochs), the model's MAE decreases significantly, indicating that the model is learning rapidly and its performance is improving markedly. As the number of training epochs increases to 200, the model reaches its lowest MAE, suggesting that it may have approached an optimal state. After 200 iterations, the MAE fluctuates slightly but remains stable overall. At specific points, it increases slightly (e.g., the MAE at 350 and 500 iterations is higher than at 200 iterations), which may be due to the model beginning to overfit the training data or the training process reaching a stable state, where further training does not result in significant performance improvements. Based on the above analysis, selecting 200 training iterations is optimal, as the model achieves a relatively low MAE value at this point.

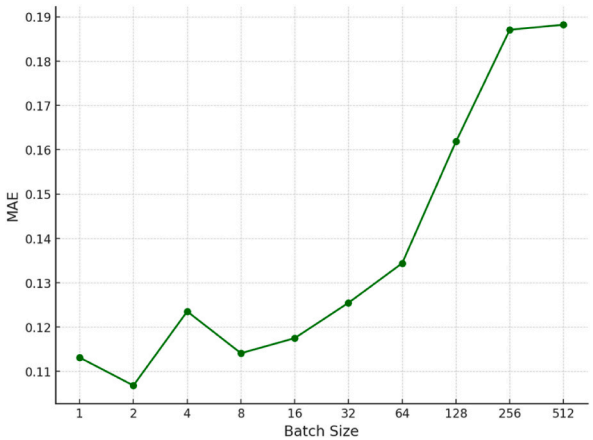


Fig. 10. Effect of batch size on model accuracy.

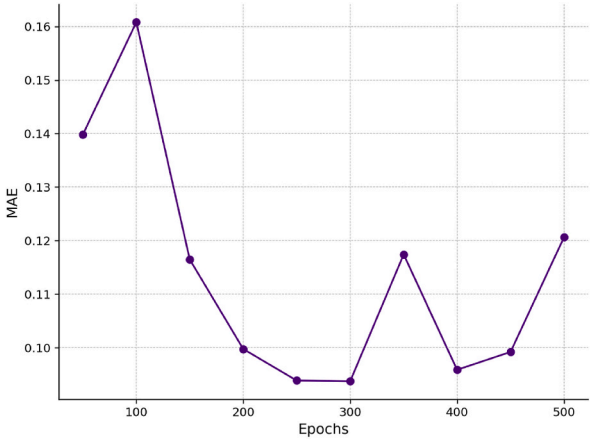


Fig. 11. Effect of epochs on model accuracy.

4. Results

4.1. System setup and operation

This research creates a prototype system of digital twins utilizing real-time observation of architectural heritage. The efficacy of the suggested five-dimensional architectural heritage digital twins model is validated through the case study of the Yingxian Wooden Pagoda in China. The system uses B/S architecture, wherein the front-end and back-end are distinct, and a defined data interface protocol facilitates efficient data transmission.

Real-time displacement monitoring data of architectural heritage components, gathered by an automated monitoring robot, is kept in a dynamic database to ensure the immediacy and correctness of the information. A displacement prediction model based on LSTM regularly assesses these time-series data. When users request monitoring and prediction data for a specific component via a web page, the system transmits requests that include the component's unique identifier (ID) to the server. The server addresses these requests by retrieving the real-time monitoring data and projected displacement trend data from the database and transmitting them to the front end using an API. The front end utilizes ECharts to render a line graph of this data, facilitating immediate and intuitive data visualization, as illustrated in Fig. 12.

4.2. Results and analysis

The geometric model that was translated underwent a thorough validation process. The validation procedure primarily emphasizes the integrity of the geometric data in the model, encompassing the precision of the model's dimensions, the accuracy of component placements, and the comprehensiveness of the connection links. According to the validation results presented in Fig. 13, the model



Fig. 12. Behavioral prediction of components.

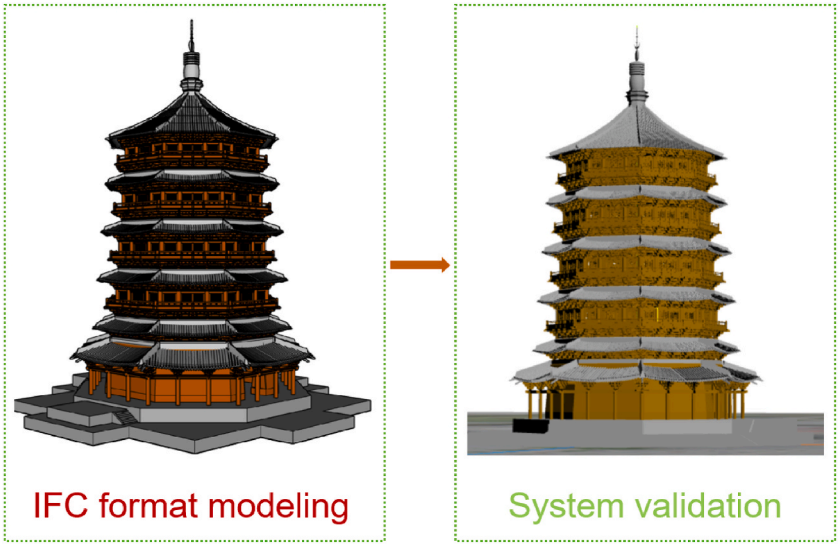


Fig. 13. Comparison of completeness of IFC format models after importing into the system.

accurately preserves all geometric information with dimensions that precisely match the real-world scenario. Additionally, the positions and connections of all components align perfectly with the design requirements. The validation process verifies the precision and usefulness of the geophysical model.

The validation of the physical model primarily emphasizes the integrity of the attribute information of the component. The digital twin system achieves the validation process by incorporating an interactive function. When the user selects a component in the system interface, the system will display the essential attribute information of the component, such as its ID and corresponding IFC (Industry Foundation Classes) type, in a small pop-up window. According to the data in Fig. 10, the validation findings indicate that all selected artefacts accurately display their attribute information, and no missing attribute information exists. This demonstrates the physical model's successful development based on its components' qualities. The translated geometric model underwent a rigorous validation process. The validation technique primarily focuses on ensuring the integrity of the geometric data in the model. This includes verifying the accuracy of the model's dimensions, the precision of component placements, and the completeness of the connection linkages. Based on the validation findings shown in Fig. 14, the model effectively maintains all geometric information with dimensions that correspond to the real-world situation. Furthermore, all components are positioned and connected per the design specifications. The validation procedure assesses the accuracy and utility of the geophysical model.

Fig. 15 illustrates the HBDT's regular model (partial) run for column M2W23. The exponential fitting model was used to fit the stress-column top offset relationship curve at the foot of the column based on the preliminary numerical simulation results, and the fitting degree reached 94 %. The above curves were corrected based on the stress values of the actual displacement data, and a polynomial fitting model was used to re-fit the column foot stress-column top offset relationship curves based on the preliminary numerical simulation results, the actual values, with a fitting degree of 97 %. The results show that the data-driven model correction method not only corrects the preliminary errors due to model simplification or parameter uncertainty but also demonstrates the accuracy of the structural linkage effects of architectural heritage.

Throughout the experiment, the Architectural Heritage Digital Twin (HBDT) framework was executed using the operational mechanism outlined in the preceding section. Fig. 16 displays the outcomes of partially executing the behavioural model of HBDT for monitoring point 223. The blue folds represent the objective observation data, the green folds represent the training data from the training set, and the red folds represent the predicted data generated by the model. The predicted time series folds in the three directions from the test set image exhibit similar high and low trends as the actual data folds. This suggests that the memory unit of the LSTM model effectively explores and comprehends the underlying patterns in the historical displacement sequence data.

Table 2 presents the outcomes of HBDT forecasts for the test set data. The root mean square error (RMSE) and the mean absolute error (MAE) reach a value of around 0.1 mm. The coefficient of determination, R2, remains constant within the range of [0.8, 1], indicating that the model exhibits a relatively high level of accuracy in predicting longitudinal, transverse, and vertical deformations. The vertical deformations are the most accurately predicted.

5. Conclusion and discussion

This study focuses on architectural legacy as the subject of research and proposes the building of a five-dimensional digital twin model. The objective is to create a highly detailed digital replica that can accurately replicate and analyze the intricate features of architectural heritage. Furthermore, the practicality and applicability of the architectural heritage digital twin 5D model framework and the architectural heritage digital twin prototype are confirmed through the use of the Yingxian Wooden Pagoda as a case study.



Fig. 14. Integrity verification of component attribute information.

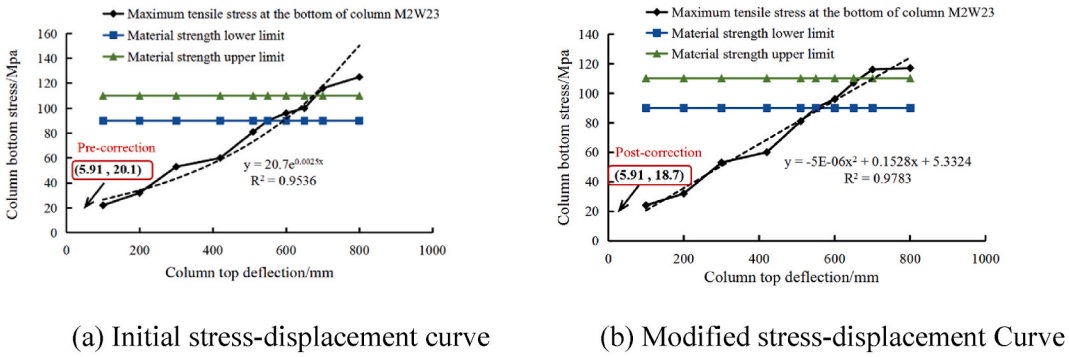


Fig. 15. Stress-displacement curve for point 223.

The following findings are derived.

(1) Constructing geometric and physical models of wooden tower buildings based on historical building information modelling

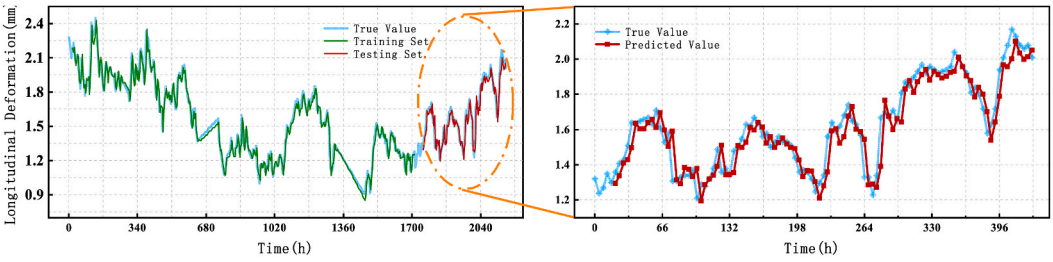
By integrating standard parametric models with irregular triangular mesh models, a detailed hybrid HBIM model of wooden tower buildings has been developed. The proposed three-step strategy—numerical model decoupling, geometric transformation, and attribute mapping—effectively ensures that the HBIM model can be seamlessly rendered and interacted with within a digital twin framework.

(2) Modeling of structural linkage rules taking into account mechanical properties

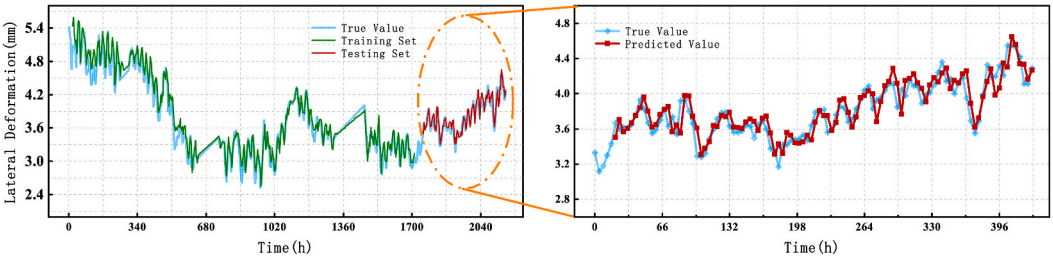
Finite element analysis was employed to examine and simulate the interactions among the components of the built heritage, informed by the mechanical properties of these components. This approach accurately represented its structural dynamics, utilizing exponential fitting models with a fit degree of 94 % and polynomial fitting models with a fit degree of 97 %. The findings indicate that employing a data-driven model correction methodology rectifies errors stemming from initial model simplification and parameter uncertainty.

(3) Construction of a prediction model for the displacement behavior of wood-frame tower buildings based on long and short-term memory networks

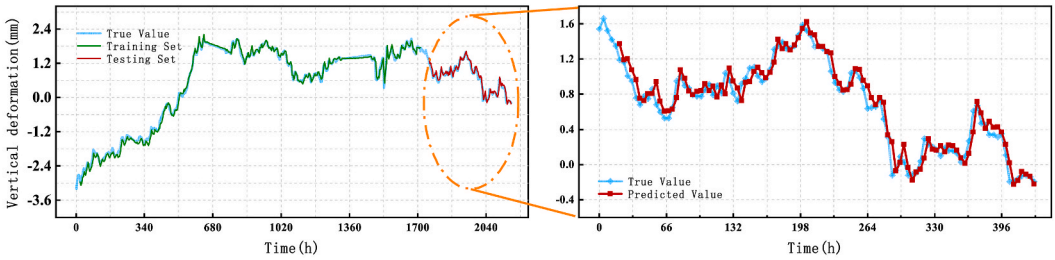
A displacement behaviour prediction model for the wooden column head of a timber tower is developed using an extended short-term memory network, enabling the forecasting of historical longitudinal, transverse, and vertical displacements for future displacements. The efficacy of the displacement behaviour prediction network is enhanced by augmenting the stacking depth of the



16(a) Longitudinal Deformation LSTM Prediction Results



16(b) Lateral Deformation LSTM Prediction Results



16(c) Vertical Deformation LSTM Prediction Results

Fig. 16. Lstm prediction results for point 223.

Table 2
Point 223 test set evaluation indicators.

223 point	RMSE(mm)	MAE(mm)	R^2
Longitudinal deformation	0.0841	0.0639	0.8834
Lateral deformation	0.1315	0.1052	0.8053
Vertical deformation	0.1130	0.0875	0.9442

LSTM. The network model is refined using "Adam + MSE," optimal hyperparameters are compared and trained, and the accuracy of the displacement prediction model is assessed through MAE, RMSE, and R2 metrics. The findings indicate that the RMSE and MAE for longitudinal, transverse, and vertical deformations converge to approximately 0.1 mm, while the coefficients of determination, R^2, stabilize within the range of [0.8, 1].

Based on research into the five-dimensional digital twin model, this study also has certain limitations, primarily manifested in the following two aspects: Although the LSTM network model demonstrates excellent predictive accuracy, careful analysis reveals that the LSTM model's predictive results exhibit significant lag. Upon further analysis, the fundamental causes of this issue may stem from two aspects: first, during training, the LSTM model optimizes by minimizing the error between the actual values and predicted values in the objective function. However, the model sometimes tends to select the closest previous observation value as the current prediction value, leading to prediction delays. Second, the feature extraction method based on a sliding time window strategy has an inherent lag, making it unable to respond promptly to trend changes in the data sequence. To address this issue, future research will focus on exploring alternative methods to mitigate prediction lag.

This study begins with the displacement prediction of ancient building structures, achieving an early assessment of structural health. However, we also recognize that displacement prediction itself can be achieved without relying on digital twins. The true advantage of digital twins lies in their ability to comprehensively and dynamically simulate and evaluate the structural response under various scenarios in a virtual environment. In future research, we will build upon the predictive results of this study to expand the functionality of the digital twin model, enabling virtual simulation and comparative analysis of repair and reinforcement strategies for ancient buildings. By simulating different repair strategies, construction techniques, and material selections within the digital twin platform, it will be possible to predict the impact of each approach on structural performance without directly intervening in fragile heritage. This will optimize the decision-making process and reduce the risks associated with on-site trials. This extension will fully leverage the potential of digital twins in the protection of ancient buildings, providing comprehensive support from monitoring and prediction to repair strategy evaluation, further enhancing the scientific and forward-looking nature of heritage preservation.

CRediT authorship contribution statement

Ming Guo: Supervision, Investigation, Data curation. **Xiaoke Shang:** Writing – original draft, Validation, Methodology. **Jiawei Zhao:** Writing – original draft, Investigation. **Yanru Shi:** Investigation. **Mengxi Sun:** Methodology. **Ming Huang:** Supervision. **Guoli Wang:** Validation, Funding acquisition. **Liqiong Liao:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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